Sai Venkata Sravan Gudepu
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PROFESSIONAL SUMMARY:
AI/ML Engineer with experience in building GenAI, Machine Learning, NLP and data driven applications. String hands-on experience in Python, SQL, FastAPI, LangGraph, RAG, LLMs, vector databases, Agentic AI workflows and cloud deployment. Experienced in developing scalable AI applications, backend APIs, and embeddings data workflows.

TECHNICAL SKILLS:
Programming Languages - Python, SQL, JavaScript
GenAI- AI Agents, LLMs, GPT, Realtime API, LangGraph, RAG, PEFT, ChromaDB, QLoRA
Other Tools – CrewAI, Azure ML, AutoGen, Hugging Face, MySQL, PostgreSQL, AWS (EC3, S3, Lambda)
Frameworks and Packages – Numpy, Pandas, FastAPI, Streamlit, OpenCV, PySpark, Tensorflow, PyTorch, Scikit-learn

EDUCATIONAL DETAILS:
Master of Science in Computer Science (2022-2024) from Central Michigan University

PROFESSIONAL EXPERIENCE:
RS Tech Inc - Dallas-TX
AI Engineer **Apr 2026 – Present**
Roles and Responsibilities:
- Designed and developed LLM powered applications using Python, LangChain and REST APIs to support enterprise automation and conversational AI use cases.
- Implemented document ingestion pipelines to extract, clean, chunk, embed, and index structured, semi-structured, and unstructured data for downstream AI applications.
- Implemented logging, monitoring, exception handling, and retry mechanisms to improve the reliability and observability of production AI services.
- Evaluated AI application performance using response accuracy, relevance, groundedness, hallucination rate, latency, and token usage metrics.
- Created reusable AI components for prompt management, model invocation, output parsing, validation, and error recovery, accelerating the development of new AI use cases.

Insight Global – AT&T, Dallas-TX
GenAI Engineer **Oct 2025 – Mar 2026**
Roles and Responsibilities:
- Worked on the development of automation, orchestrating specialized AI Agents with state management and conditional routing.
- Improved LLM response reliability by building a pgvector based RAG pipeline that retrieved enterprise knowledge before generation, reducing unsupported responses and grounding answers in internal business context.
- Made an impact by implementing resume/pause capabilities using LangGraph's in-built primitive for sensitive operations.
- Developed AI applications with Python, FastAPI, REST APIs and webhook based workflows, enabling real-time interaction between LLM systems and enterprise platforms.
- Automated manual workflow steps by building LangGraph AI agents with retrieval, tool calling, validation, routing, and human approval checkpoints.
- Contributed to the framework by reducing latency approximately by 5 secs of workflow of agents.
- Designed human-in-the-loop (HITL) checkpoints within agentic AI workflows, enabling approval gates, escalation paths, and manual review for necessary sensitive actions.
- Used AI assisted development tools such as Cursor and GitHub Copilot to accelerate coding, debugging, refactoring, test generation, and documentation while validating code quality.
- Designed advanced prompt strategies including system prompts, few shot examples, instruction following patterns, intent classification prompts, and structured output prompts to improve LLM reliability.
- Built and maintained Pydantic models in LLM powered FastAPI applications to validate API request/response schemas, enforce data types, handle nested payloads, and improve reliability of backend AI services.

Insight Global – Microsoft, Dallas-TX
AI Engineer **June 2025 – Oct 2025**
Roles and Responsibilities:
- Reduced manual data validation effort by approximately 70% by building AI powered validation workflows that detected inconsistencies, missing values, and anomaly patterns before SME review.
- Built and hosted scalable web applications using Streamlit and Flask, enabling seamless access to AI workflows.
- Delivered AI powered MVPs using Azure OpenAI, and CrewAI, helping SMEs validate GenAI use cases faster.
- Cross collaborated with stakeholders to gather requirements and deliver AI applications tailored to business needs.

- Implemented vector search using ChromaDB, and FAISS, improving semantic retrieval grounding LLM responses in trusted enterprise knowledge.
- Contributed to delivering the requirements by solving complex problems in automation pipelines, and accelerating project timelines.
- Utilized Azure DevOps to streamline the software development lifecycle.

QuantumVerse AI – Seattle, WA
AI Engineer **Jan 2024 - June 2025**
Roles and Responsibilities:

- Improved prompt consistency across 7 AI Agent tasks by defining reusable prompt templates with role instructions, structured outputs and validation rules.
- Built and tested RAG and PEFT based LLM workflows to improve domain specific response relevance, combining retrieval grounding, prompt tuning, and lightweight model adaptation.
- Developed ML/NLP pipelines supporting business workflows, which includes semantic retrieval, summarization, and AI assisted automation.
- Integrated advanced AI tools/frameworks including CrewAI, LiveKit and others to support real-time agent interaction and workflow automation.
- Developed and validated a Proof of Concept (PoC) for assessing AI agent feasibility in automating business operations, contributing to potential revenue generation.
- Owned the end-to-end ML lifecycle for GenAI and ML systems, including data exploration, feature engineering, model selection, training, validation, deployment and iterative optimization.
- Performed data wrangling by cleaning, normalizing, and transforming raw data from multiple sources (SQL tables, JSON files) using Python and Pandas, preparing datasets for analytics and machine learning models.
- Applied NLP and reinforcement learning techniques to improve model performance and user query understanding.
- Utilized Azure DevOps and CI/CD pipelines to streamline the software development lifecycle.

Dhanush Infotech Pvt.Ltd - Hyderabad, India
Python Developer **Jan 2021 – June 2022**
Roles and Responsibilities:

- Reduced ML pipeline processing time by 50% by optimizing Python data pipelines, improving transformation logic, and reducing redundant processing steps.
- Experienced with Python's object-oriented programming and also worked on divergent formats of data like CSV, JSON, YAML and XML.
- Performed EDA to extract meaningful insights and prepare high-quality features for ML workflows.
- Visualized analytical outcomes using Python based tools (Matplotlib, Seaborn) for stakeholder reporting.
- Conducted advanced statistical analyses such as regression, hypothesis testing, and time-series forecasting to uncover patterns and evaluate model performance.
- Improved model selection reliability by evaluating several ML models using accuracy, precision, recall, F1-score, ROC-AUC, and RMSE across classification and regression use cases.
- Engineered model features from raw structured and semi structured datasets using Python, Pandas, NumPy, and SQL, improving model training quality and analytical reliability.
- Applied NLP techniques including text preprocessing, tokenization, vectorization, keyword extraction, and sentiment analysis to derive insights from unstructured text data.
- Built machine learning models using Scikit-learn, TensorFlow, and PyTorch, supporting classification, regression and recommendation use cases.
- Used SQL to perform data validation, anomaly checks, and exploratory data analysis, ensuring data quality and consistency prior to model development and reporting.
- Worked cross-functionally with software engineers, data analysts, and product managers to integrate ML models with business applications.
- Created and documented data validation processes and business rules to ensure high integrity in analytics reports.